

Problem 2

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Problem 1 The velocity of an object moving along a straight line is given by $v(t)$ (in ft/s) and we only know the following: $\int_0^3 v(t) dt = -3$, and $\int_3^4 v(t) dt = 5$.

Compute the following displacements, if possible. If it is not possible, give examples explaining why not.

- (a) $s(3) - s(0)$
- (b) $s(4) - s(0)$
- (c) Find the displacement during the interval $[0, 4]$ if the velocity at the time t , $0 \leq t \leq 4$, was $v(t) + 2$ ft/s instead?
- (d) Find the displacement during the interval $[0, 4]$ if the velocity at the time t , $0 \leq t \leq 4$, was $5v(t)$ ft/s instead?